

15.1 图 15.1 所示电路，电源频率 $f = 50\text{Hz}$ ， $R_L = 40\Omega$ 。(1) 要求输出电压 $U_{O(AV)} = 100\text{V}$ ，求变压器副边电压的有效值 U_2 ，并确定二极管的参数。(2) 接入滤波电容 C (虚线处)，若要求满足 $R_L C = 2.5T$ ，求 C 的值。若仍要求输出电压 $U_{O(AV)} = 100\text{V}$ ，求接入滤波电容后 U_2 。

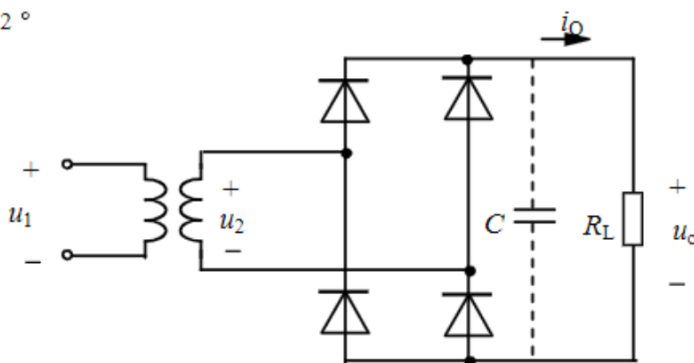


图 15.1

$$(1) U_{O(AV)} \approx 0.9 U_2 \Rightarrow U_2 \approx 111.1\text{V}$$

$$I_D(AV) = \frac{U_{O(AV)}}{2R_L} \approx \frac{100}{2 \times 40} = 1.25\text{A}$$

$$I_F = 3 I_D(AV) = 3.75\text{A}$$

$$U_{DRM} = \sqrt{2} U_2 = 157.119\text{V}$$

$$U_{RM} = 2 U_{DRM} = 314.238\text{V}$$

$$(2) f = 50\text{Hz}, T = \frac{1}{f} = 0.02\text{s}$$

$$C = \frac{2.5 T}{R_L} \approx \frac{2.5(0.02)}{40} = 1.25\text{mF}$$

$$U_{O(AV)} \approx 1.2 U_2 \quad U_2 = 83.333\text{V}$$

15.4 Determine the value of the potentiometer resistance R_W and the value of the zener voltage U_Z for the series regulator in Figure 15.4, if the output voltage V_{OUT} can be regulated from 1.6V to 20.8V.

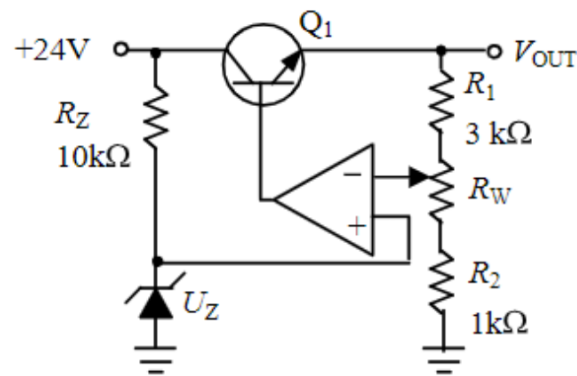


Figure 15.4

$$V_{out\ max} = 20.8\ V = \frac{R_1 + R_2 + R_W}{R_2} U_Z$$

$$V_{out\ min} = 1.6\ V = \frac{R_1 + R_2 + R_W}{R_2 + R_W} U_Z$$

$$20.8 (1000) = (1000 + R_W) U_Z = 1.6 (1000 + R_W)$$

$$R_W = 12\ k\Omega \quad U_Z = 1.3\ V$$